GERBE PROPERTY IS INVARIANT UNDER EQUIVALENCE OF CATEGORIES OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. The converse direction is the proven direction instantiated on the reversed equivalence data; the source does not spell it out and no omission is declared.

Allowed standard prerequisites (editorial).

- Stack in groupoids invariance. lemma-stack-in-groupoids-equivalent (same chapter, named). Transfers the stack-in-groupoids part of the gerbe definition along the equivalence.
- Fibrewise equivalence. categories-lemma-equivalence-fibred-categories (cross-chapter, named). Transfers the isomorphism G(x)|_{U_i} = G(y)|_{U_i} back to x|_{U_i} = y|_{U_i} in S_2.

Source reference. Original reference 003Z: focused-seeds/geometry/S9-stk-gerbe-equivalent/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 042X: sources/stacks-stacks/pinned-original-source.tex, source lines 1207–1207. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{S}_1$, $\mathcal{S}_2$ be categories over $\mathcal{C}$. Suppose that $\mathcal{S}_1$ and $\mathcal{S}_2$ are equivalent as categories over $\mathcal{C}$. Then $\mathcal{S}_1$ is a gerbe over $\mathcal{C}$ if and only if $\mathcal{S}_2$ is a gerbe over $\mathcal{C}$.*

Proof. Assume $\mathcal{S}_1$ is a gerbe over $\mathcal{C}$. By Lemma 042X we see $\mathcal{S}_2$ is a stack in groupoids over $\mathcal{C}$. Let $F : \mathcal{S}_1 \to \mathcal{S}_2$, $G : \mathcal{S}_2 \to \mathcal{S}_1$ be equivalences of categories over $\mathcal{C}$. Given $U \in \mathrm{Ob}(\mathcal{C})$ we see that there exists a covering $\{U_i \to U\}$ such that $(\mathcal{S}_1)_{U_i}$ is nonempty. Applying F we see that $(\mathcal{S}_2)_{U_i}$ is nonempty. Given $U \in \mathrm{Ob}(\mathcal{C})$ and $x, y \in \mathrm{Ob}((\mathcal{S}_2)_U)$ there exists a covering $\{U_i \to U\}$ in $\mathcal{C}$ such that $G(x)|_{U_i} \cong G(y)|_{U_i}$ in $(\mathcal{S}_1)_{U_i}$. By Categories, Lemma 003Z this implies $x|_{U_i} \cong y|_{U_i}$ in $(\mathcal{S}_2)_{U_i}$. □

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